

Canadian Building Permits Live Tracker

A plain-language walkthrough of the pipeline — what each step does and why, written for the first time you're pulling live government data.

Project	StatCan Building Permits Live Tracker
Files	StatCan_Building_Permits_Live_Tracker.ipynb, statcan_sql_setup.sql, this PDF
Data source	Statistics Canada, Table 34-10-0066-01 (Web Data Service API)
Stack	Python (Jupyter) → pandas/statsmodels/scikit-learn → SQL → Power BI

1. Why StatCan needs no API key

Statistics Canada's Web Data Service (WDS) is a fully open public API — no signup, no key, no authentication headers. That makes this project a good complement to the AESO one: AESO taught you the key/header pattern that most commercial APIs use, and this one shows you the simpler, fully-open pattern that many government and public-data APIs use instead.

The trade-off: because anyone can call it, there's no per-user rate limit protecting you — StatCan just asks that you be reasonable with request frequency, which a monthly refresh easily satisfies (this table itself is only updated monthly).

2. The two-step download pattern

Unlike AESO (one call, get your data back directly), StatCan's full-table download works in two steps — a pattern that shows up constantly in real APIs:

- **Step 1:** call the API and ask for a download link for the table you want
- **Step 2:** the API replies with a URL pointing to a ZIP file — you then download and unzip that file yourself

```
GET https://www150.statcan.gc.ca/t1/wds/rest/getFullTableDownloadCSV/34100066/en
→ {"status": "SUCCESS", "object":
  "https://www150.statcan.gc.ca/n1/tbl/csv/34100066-eng.zip"}
```

This "ask for a link, then fetch it" pattern exists because the actual file is large and StatCan generates/hosts it separately from the API that answers your request. You'll see the same two-step shape with cloud storage exports, reporting tools, and many enterprise APIs — recognizing it here means you'll recognize it again.

3. What's actually in the table

Table 34-10-0066-01 ("Building permits, by type of structure and type of work") is StatCan's monthly release of building permit values across Canada, broken down by province, census metro area, structure type (residential/non-residential), and type of work. The full table is large — every region and category StatCan tracks — so the notebook filters it down immediately to five regions (Canada, Alberta, Ontario, British Columbia, Quebec) and the residential/non-residential split, which is enough detail for a regional trend dashboard without the noise of every census subdivision.

4. Three analysis techniques, and why each one

Year-over-year comparison (pandas)

The simplest and most requested comparison in economic reporting: this month vs. the same month last year. It's a single line of pandas (`pct_change(periods=12)`) but it's the number every construction-sector stakeholder asks for first.

Seasonal decomposition (statsmodels)

Construction activity dips every winter and rises every spring — that's not a trend, it's a known seasonal pattern. Seasonal decomposition splits the series into three parts: **trend** (the underlying long-run direction), **seasonal** (the repeating yearly pattern), and **residual** (whatever's left over — the genuine surprises). This matters because without it, a January dip could be mistaken for a slowdown when it's actually just winter.

Linear trend forecast (scikit-learn)

A simple linear regression fit to the last 24 months of the trend gives a 3-month forward projection. This is intentionally simple — not a sophisticated forecasting model — because the goal is a defensible, easy-to-explain directional estimate, which is what's actually asked for in a monthly reporting cycle, not a black-box prediction nobody can interpret.

5. Exporting to SQL and Power BI

The notebook writes three tables into a SQLite database: `building_permits` (the cleaned raw data), `national_trend` (YoY and rolling averages), and `forecast` (the 3-month projection). Keeping these separate means each one maps directly to a different type of chart in Power BI — detail table, trend line, and forecast overlay.

- Open Power BI Desktop → **Get Data** → **Text/CSV** → select `statcan_building_permits.csv` for the regional detail
- For the trend and forecast tables, use **Get Data** → **ODBC** with a SQLite driver pointed at `statcan_building_permits.db`, or export those tables to CSV the same way
- Build: a regional trend line chart, a YoY % change KPI card, and a forecast overlay on the national trend

See `statcan_sql_setup.sql` for the full schema and ready-to-run queries, including a regional ranking query and a forecast-vs-actual comparison you can re-run once new live months arrive.

6. Making it live and recurring

StatCan releases this table monthly (a given month's permits are typically published 5-6 weeks later). To keep the tracker current automatically:

- Set `USE_LIVE_API = True` in Section 2 — no key needed, it pulls immediately
- **Windows:** Task Scheduler → run `jupyter nbconvert --to notebook --execute --inplace StatCan_Building_Permits_Live_Tracker.ipynb` monthly
- **Mac/Linux:** the same command via `cron`, scheduled for early each month
- Turn on Power BI's **Scheduled Refresh** so the dashboard reflects each new pull automatically

*Companion files: `StatCan_Building_Permits_Live_Tracker.ipynb` (the pipeline) and `statcan_sql_setup.sql` (schema + queries). For a look at how an analyst would reason through this problem end-to-end — not just how the code works — see *Analyst_Thought_Process.pdf*.*